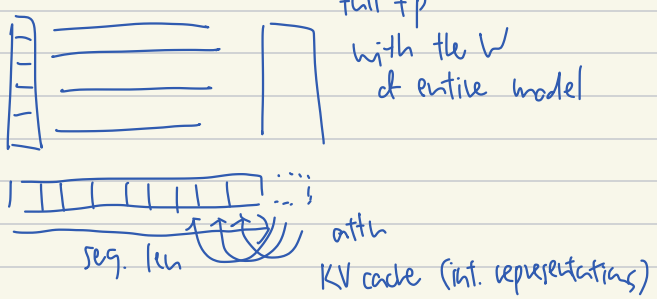


How batch size affects token cost and speed

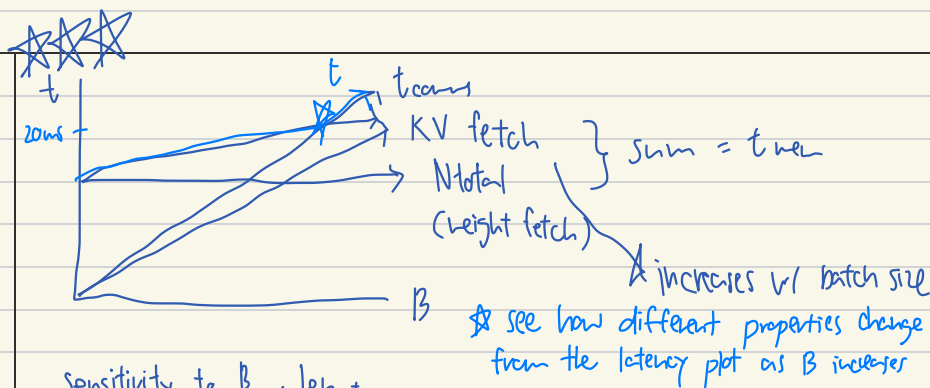
24/9/2016

Big effect	Batch size
2 principles of analysis	Roofline analysis mem. bw cap. perf. Weights KV cache $t \geq \max(t_{mem}, t_{compute})$ $t_{compute} = \text{mhl all active params}$ $\star$ caveat: didn't count attention $\frac{B \cdot N_{active}}{FLOPs}$ $t_{mem} \geq \frac{N_{total} + B \cdot \overset{\text{spare attn}}{\text{len ctx}} \cdot \text{Bytes/token}}{\text{memory bytes/s}}$
N is params	
autoreg. inf.	
	full fp with the W of entire model  seq. len attn KV cache (int. representations)

Use attention to look back at prev. tokens thru KV cache

KV cache $\propto B \cdot \text{ctx len}$

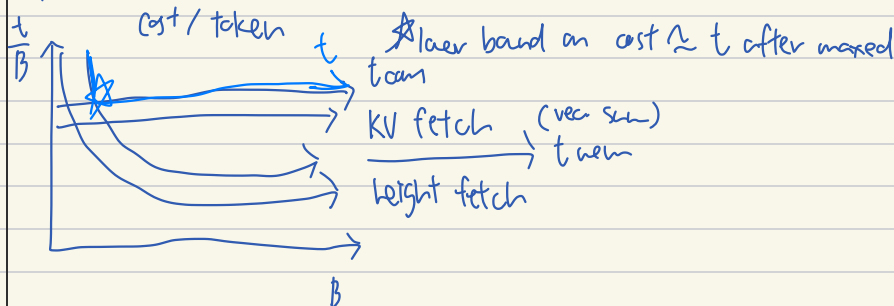
Latency plot



★ same inter. point

Sensitivity to B , len ctx

KV fetch varies w/ ctx len
particular len. both com/mem bound $\approx$ ideal



mem. = model N
den. = hardware N

$$\frac{B \cdot N_{active}}{FLOPs} = \frac{N_{total}}{\text{mem bytes/s}}$$

fp4

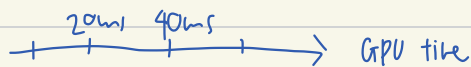
$$B \geq 300 \times \frac{8}{\text{sparsity}}$$

equil $\iff \frac{FLOPs}{\text{mem BW}} = B \cdot \frac{N_{active}}{N_{total}} \approx 300$

w ≈ 2 or ≈ 3

optimal B size should $\uparrow$

amortization = cost of loading weights spread across generating a lot of tokens
 $B \uparrow$ reduce cost/token



worst latency is 40ms

B depends on sparsity

ideal B size ≈ 2000

think of it as
users
or tokens
per second

$$\text{tok/s} = B/t = 15\text{ms} = 64B \sim 128\text{k tok/s}$$

more sparsity less compute

batch size $\uparrow$ compute bound

$$\frac{E}{K} \downarrow = \text{sparsity} \uparrow$$

fewer active params. Model performance $\downarrow$

lots of users, get a very sparse model

we want

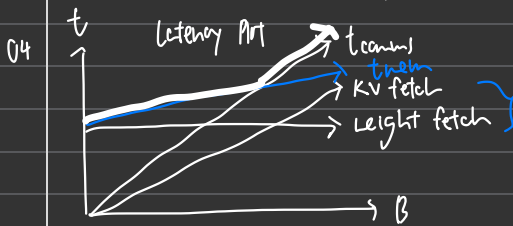
$t_{\text{com}} \downarrow$ sparsity $\uparrow$ $B \uparrow$ mem capacity $\uparrow$

How batch size affects token cost and speed

01 $t \geq (t_{mem}, t_{compute})$

02 $t_{compute} = \frac{B \cdot N_{active}}{FLOPs}$

03 $t_{mem} \geq \frac{N_{total} + B \cdot ctx_{len} \cdot KV \text{ Bytes token}}{\text{memory bw}}$



05 lower bound = $t \geq (t_{mem}, t_{compute})$ bcz even if you spent very little time on compute by processing less sequence, you still need to load the weights.

06 $t_{compute}, KV \text{ fetch}$

07 ≈ 300

08
$$\frac{B \cdot N_{\text{active}}}{\text{FLOPs}} = \frac{N_{\text{total}}}{\text{mem BW}}$$

$$B = \frac{\text{FLOPs}}{\text{mem BW}} \cdot \frac{N_{\text{total}}}{N_{\text{active}}}$$
$$= 300 \cdot \text{sparsity}$$

$$B \geq 300 \cdot \text{sparsity}$$

09
$$\text{HBM drain time} = \frac{\text{mem C}}{\text{mem BW}}$$

It is impossible for the HBM to load all the weights that fast

Creates idle time for FLOPs